THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The value of $24^2 - 23^2$ is: 1

(a) 24 **(b)** 47 **(c)** 49 **(d)** 48

2. The sides of two squares are in the ratio 2 : 3. The ratio of their areas is: 1

(a) 4 : 3 **(b)** 2 : 3 **(c)** 4 : 9 **(d)** 2 : 9

3. If the difference between the CI and the SI on a sum for 2 years at 4% is ₹ 12, the sum (in ₹) is: 1

(a) 6750 **(b)** 15000 **(c)** 7500 **(d)** 8250

4. A ladder 65 m long reaches 63 m up a wall. Its top slides down 3 m along the wall. The foot of the ladder moves away from the wall by (in m): 1

(a) 25 **(b)** 9 **(c)** 16 **(d)** 3

5. The wages of 12 workers for 20 days are ₹ 36000. The wages of 9 workers for 14 days at the same daily rate (in ₹) are: 1

(a) 36000 **(b)** 18900 **(c)** 27000 **(d)** 19050

6. $(a+b)^2 = a^2 + b^2$ is true only when: 1

(a) $a = 1$ **(b)** $ab = 0$ **(c)** $a = b$ **(d)** $a + b = 0$

7. The number x for which 4 : 9 = 12 : x is: 1

(a) 18 **(b)** 25 **(c)** 21 **(d)** 27

8. A sum triples in 4 years at simple interest. The rate of interest per annum is: 1

(a) 55% **(b)** 50% **(c)** 100% **(d)** 45%

9. The legs of a right triangle are in the ratio $8 : 15$ and its hypotenuse is 51 cm. Its perimeter (in cm) is: 1

(a) 69 **(b)** 120 **(c)** 123 **(d)** 102

10. The present ages of Ria and Sia are in the ratio 2:5. After 9 years the ratio will be 7:13. Ria's present age (years) is: 1

(a) 21 **(b)** 12 **(c)** 10 **(d)** 30

11. If $x + y = 11$ and $xy = 28$, then $x^2 + y^2$ equals: 1

(a) 65 **(b)** 121 **(c)** 93 **(d)** 177

12. If x : y = 3 : 4 and y : z = 6 : 5, then x : y : z is: 1

(a) 3 : 10 : 5 **(b)** 18 : 24 : 20 **(c)** 9 : 12 : 10 **(d)** 3 : 4 : 5

13. **Assertion (A):** The ratio $15 : 25$ in its simplest form is $3 : 5$. 1

Reason (R): The ratio of two quantities of the same kind has no unit.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. **Assertion (A):** A shirt marked at ₹800 is sold at a 10% discount; its selling price is ₹720. 1

Reason (R): Profit or loss percent is calculated on the cost price.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. **Assertion (A):** $(5, 12, 13)$ is a Pythagorean triplet. 1

Reason (R): $(3, 4, 5)$ is a Pythagorean triplet.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. If $(x + p)(x + q) = x^2 + 8x + 15$, find $p^2 + q^2$. 2

17. Divide ₹ 1200 into parts proportional to $\frac{1}{2}$ and $\frac{1}{3}$. 2

18. Rahul invests ₹ 2500 at 20% p.a. compounded annually. Find the amount after 2 years and write it as a fraction of the principal. 2

19. A kite string 50 m long is stretched straight and the kite is 48 m above the boy's hand. Find the horizontal distance of the boy from the point directly below the kite. 2

20. The present ages of a father and son are in the ratio 7:2. After 6 years the ratio will be 8:3. Find their present ages. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. In a pie chart, the sector for Science is 40° and represents 12 students, while the sector for Maths is 80°. Find the total number of students and the number who like Maths. 3

22. If $x - y = 6$ and $xy = 40$, find $x^2 + y^2$ and $(x + y)^2$. 3

23. The ratio of the present ages of a father and son is 5 : 2. After 6 years the ratio will be 2 : 1. Find their present ages. 3

24. A merchant sells two items at ₹ 2640 each, one at a gain of 20% and the other at a loss of 20%. Find the overall gain or loss in rupees. 3

OR

A ladder 53 m long, with its foot in a street, reaches a window 45 m high on one side. Without moving its foot it is turned to reach a window 28 m high on the opposite side. Find the width of the street. 3

25. A garrison of 180 men has provisions for 40 days. After 10 days 45 men leave. For how many more days will the remaining provisions last for the men who stay? 3

OR

Simplify $(4x + 5)^2 - (4x - 5)^2$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Arjun buys a jacket with a marked price of ₹2,500 at a 20% discount. GST of 12% is then added on the discounted price. 5

(i) Find the discount.

(ii) Find the price after the discount.

(iii) Find the GST amount.

(iv) Find the final amount Arjun pays and by how much it is less than the marked price.

27. The sides of a rectangle are $(4x + 3)$ m and $(4x - 3)$ m. Find its area as an expression, its value at $x = 6$, and by how much it is less than the area of a square of side $4x$ m. 5

OR

A, B and C invest ₹ 3000, ₹ 2000 and ₹ 4000 for 4, 6 and 2 months respectively. A profit of ₹ 1200 is shared in the ratio of (investment $\times$ time). Find the ratio and each share. 5

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND